

RESEARCH ARTICLE

Time series approaches for air pollution forecasting in Çanakkale: A performance comparison of ETS and Box-Jenkins models

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ABSTRACT

Air pollution is a complex process that requires statistical modeling and forecasting due to the temporal variability in the concentrations of different pollutant components in the atmosphere. This study analyzes monthly concentrations of PM₁₀, PM_{2.5}, SO₂, NO₂, and O₃ measured in Çanakkale between 2019 and 2024 using time series methods to generate future forecasts. Factor analysis was first applied to identify the underlying structure among pollutants and to select the most influential variables for modeling; as a result, PM₁₀ and SO₂ were found to load most strongly on the primary factor associated with air pollution in the region. For these two pollutants, which were examined in terms of stationarity, trend, and seasonal components, forecasting models were developed based on Exponential Smoothing techniques (Simple, Holt, Damped, Holt-Winters additive and multiplicative) and the Box-Jenkins methodology (ARIMA, SARIMA). Model performance was evaluated using RMSE, MAE and MAPE for both in-sample and out-of-sample periods. The Holt-Winters multiplicative model achieved the highest forecasting accuracy, with MAPE values of 11.040% for PM₁₀ and 20.253% for SO₂ by effectively capturing trend and proportional seasonality. This study is distinctive in that it provides a comparative evaluation of ETS and Box-Jenkins approaches for multiple air pollutants and develops region-specific forecasting models for Çanakkale, thereby offering novel methodological and practical insights for air quality assessment in Türkiye.

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INTRODUCTION

Air pollution is recognized as one of the most critical environmental issues worldwide due to its adverse impacts on human health and ecosystems. According to the World Health Organization, pollutants such as particulate matter (PM), nitrogen dioxide (NO₂), sulfur dioxide (SO₂), carbon monoxide (CO), and ozone (O₃) are among the leading causes of respiratory and cardiovascular diseases, contributing to millions of premature deaths [1]. Air quality is not only a key determinant of public health but also plays a significant role in climate change, agricultural productivity, and ecosystem balance [2]. Therefore, modeling pollutant concentrations and forecasting their future levels is of paramount importance for both scientific research and environmental planning.

In Turkey, air quality management is regulated under the Air Quality Assessment and Management Regulation, which sets limit values aimed at protecting public health. Çanakkale is among the regions requiring regular monitoring due to emissions from industrial activities, heating, and traffic, as it is strategically located at the intersection of road, air, and sea transport between the Marmara and Aegean regions and

hosts numerous industrial facilities operating in sectors such as food, furniture, metal, chemical, plastic, tobacco, and machinery manufacturing. These characteristics make Çanakkale a strategically important area where both regional transport of pollutants and local emission sources interact, and therefore a suitable case study for air pollution modeling. Data from the Marmara Clean Air Center and Continuous Monitoring Stations in Çanakkale indicate periodic increases in parameters such as PM₁₀ and NO_x [2, 3].

Among the pollutants examined in this study, PM₁₀ and PM_{2.5} belong to the particulate matter class and pose significant health risks to the respiratory system. NO₂ formed primarily from fossil fuel combustion, are nitrogen oxide compounds that play an important role in atmospheric chemistry and the formation of secondary pollutants. SO₂ is mainly released during coal-based energy production and industrial activities and is a major contributor to acid rain. O₃, when formed at the tropospheric level, acts as a harmful pollutant and tends to increase during summer months due to photochemical reactions [4, 5]. The concentrations of these pollutants are strongly influenced by meteorological factors such as temperature, humidity, pressure, wind speed, precipitation, and sunshine duration. Since air pollution can result from

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multiple factors, it is essential to examine and analyze the origins of air pollutants and the relationships among them [6].

Recent studies on air pollutants in Çanakkale province were also examined. For the Çanakkale region, Aydın and Mentese [7] investigated air quality in the Gelibolu district based on VOC measurements and examined both spatial and temporal (daily and seasonal) variations, showing that routine road traffic and ferry-related vehicle loading substantially increased VOC levels. In addition, Akyürek [8] analyzed the forest fire that occurred in Çanakkale between 22–24 August 2023 using Sentinel-2 and Sentinel-5P satellite images and Geographic Information Systems, and reported that the fire affected an area of 4191.93 hectares and significantly increased atmospheric NO₂ concentrations, thereby deteriorating air quality. At the local scale, these recent studies have mainly focused on monitoring air quality, mapping its spatial distribution and identifying emission sources; by contrast, studies providing a comprehensive framework based on statistical time series modelling and forecasting remain relatively scarce.

In the literature on air pollution forecasting, both classical time series models and artificial intelligence-based approaches have been employed. The Box-Jenkins methodology (ARIMA, SARIMA) is widely recognized as a robust statistical approach for modeling series with trend and seasonality [9]. Existing literature shows that classical time series models such as ARIMA and SARIMA are widely applied for forecasting environmental and climate-related variables. Several studies have demonstrated the effectiveness of SARIMA models in capturing seasonality and improving forecast accuracy for climate variables such as temperature and precipitation [10]. Temiz and Turgut [11] performed a time series analysis of PM₁₀ concentrations in Ankara using the Box-Jenkins methodology and identified an ARIMA (3,0,0) model as the most appropriate specification for forecasting future PM₁₀ levels. In a comprehensive national-level study, Cekim [12] compared ARIMA, ETS, and Singular Spectrum Analysis (SSA) models to forecast PM₁₀ concentrations across Turkey's most polluted cities. The results demonstrated that while ETS provided satisfactory forecasts in some regions, non-parametric SSA achieved the lowest overall prediction errors, highlighting the importance of comparing traditional and state-space approaches in air quality modeling.

Exponential smoothing methods, such as Holt-Winters, have also proven effective, particularly for short- and medium-term forecasts [13]. For instance, Wu et al. [14] applied the Holt-Winters method for predicting the Air Quality Index in China and demonstrated its capability in capturing seasonal components. Similarly, Setiawan [15] utilized ARIMA for PM forecasting in Madrid, while Hansun et al. [16] implemented modified exponential smoothing techniques for Jakarta. Studies conducted in Turkey have generally focused on either a single pollutant or a single modeling approach [4, 5, 17, 18]. Comprehensive studies that simultaneously compare ETS and ARIMA methods across multiple pollutants remain limited in the existing literature.

The objective of this study is to model the major air pollutants measured in Çanakkale (PM₁₀, PM_{2.5}, SO₂, NO₂, and O₃) for the period 2019–2024 using time series methods and to produce forecasts for future periods. For this purpose, the study was organized as follows: First, time series plots of air quality variables were generated, and the variables were descriptively analyzed. Next, factor analysis was applied to identify the most influential pollution parameters in Çanakkale. Based on the results of the factor analysis, selected

pollutants were modeled using both Exponential Smoothing and Box-Jenkins methodologies. In this context, the central research question of the study is whether, owing to their more flexible treatment of trend and especially seasonal components, ETS models outperform Box-Jenkins models in forecasting the air pollutions. The performance of the resulting models was then evaluated using various error metrics to identify the best-performing models. This study is distinctive in that it comparatively examines multiple pollutants using both ETS and Box-Jenkins approaches and provides forecasts for the Çanakkale region—a topic rarely addressed in the Turkish literature. In this context, the research stands out as one of the first comprehensive studies to compare the performance of two different classical time series approaches in detail.

MATERIALS AND METHODS

In this study, future forecasts were generated using major air pollutants (PM₁₀, PM_{2.5}, SO₂, NO₂, and O₃) measured in Çanakkale between 2019 and 2024. The forecasting process involved a comparative application of two classical time series approaches: ETS methods and the Box-Jenkins methodology. During the modeling phase, the stationarity of the series was examined using the Augmented Dickey-Fuller (ADF) test, and trend and seasonality components were analyzed. The predictive performance of the models was evaluated using metrics such as RMSE, MAE and MAPE.

All statistical analyses and forecasting procedures in this study were conducted using R statistical software (RStudio IDE, version 2024.04.1 Build 748), with relevant packages for time series modeling and visualization, including *forecast* (for *auto.arima()*, *ets()*, and *accuracy()* functions), *tseries* (for *adf.test()*) and *ggplot2* (for graphical representation of time series and forecasts).

Forecasting with Exponential Smoothing Methods

Exponential smoothing methods are highly effective and widely used approaches for forecasting in time series analysis. The fundamental principle of these methods is to assign greater weights to more recent observations in the series when predicting future values. In this way, the most recent data points are considered more significant, thereby improving forecast accuracy. The main types of exponential smoothing methods include: Simple Exponential Smoothing (SES), Holt's Exponential Smoothing, Damped Exponential Smoothing, and Holt-Winters Exponential Smoothing.

Simple exponential smoothing, SES

SES is used when the time series does not contain components such as trend or seasonality. In other words, the goal of this method is to predict the current level of the time series in cases where there is no general trend or seasonal effect. For future forecasts, greater weight is assigned to the most recent observations of the series.

In a time series with T observations, let the observed value at time t be y_t . The forecast for h periods ahead made at time t is given by:

$$\text{Forecast Equation: } \hat{y}_{t+h|t} = \ell_t$$

$$\text{Level Equation: } \ell_t = \alpha y_t + (1 - \alpha)\ell_{t-1}$$

The initial forecast value is taken as the arithmetic mean of the series. Here, α is the smoothing parameter, which takes a value between 0 and 1, and its optimal value is determined by minimizing the mean squared error (MSE).

Holt exponential smoothing

If the time series to be forecasted contains only a trend component, this method is preferred. In addition to the Simple Exponential Smoothing model, a trend term is included in the forecasting model since the data exhibit a trend. This method uses two smoothing parameters, both ranging between 0 and 1, which are determined by minimizing the mean squared error [19]. To enable forecasting of trend-containing data, Holt extended the simple exponential smoothing method as follows:

Forecast Equation: $\hat{y}_{t+h|t} = \ell_t + hb_t$

Level Equation: $\ell_t = \alpha y_t + (1 - \alpha)(\ell_{t-1} + b_{t-1})$

Trend Equation: $b_t = \beta(\ell_t - \ell_{t-1}) + (1 - \beta)b_{t-1}$

Here, α and β are the smoothing parameters, and b_t represents an estimate of the trend (slope) at time t . In this model, α controls the smoothing of the level, while β controls the smoothing of the trend. Both parameters are optimized within the range (0,1) by minimizing the mean squared error [20]. The initial values for level and trend are typically obtained using parameter estimates from a simple linear regression applied to the series [21].

Damped exponential smoothing

Proposed by Gardner and McKenzie [22], damped exponential smoothing extends Holt's method by introducing a damping parameter that gradually reduces the influence of the trend over time. This approach allows for more stable and reliable forecasts in the long term. Particularly for multiplicative models, where the trend is expected to diminish or exhibit dynamic changes over time, the damped method often produces more robust results [23]. In addition to the smoothing parameters, this method incorporates a damping parameter, θ , which lies within the interval $0 < \theta < 1$ [24].

Forecast Equation: $\hat{y}_{t+h|t} = \ell_t + (\theta + \theta^2 + \dots + \theta^h)b_t$

Level Equation: $\ell_t = \alpha y_t + (1 - \alpha)(\ell_{t-1} + \theta b_{t-1})$

Trend Equation: $b_t = \beta(\ell_t - \ell_{t-1}) + (1 - \beta)\theta b_{t-1}$

It is evident that when $\theta = 1$, the model reduces to Holt's linear exponential smoothing method.

Holt-Winters exponential smoothing

The Holt-Winters method is an extension of the simple and Holt exponential smoothing methods, specifically designed for time series with seasonality. To account for seasonal effects, this method can be applied using two different models depending on the structure of the series: additive and multiplicative seasonal models. In the additive model, seasonal fluctuations remain constant regardless of the overall level of the series, while in the multiplicative model, the magnitude of seasonal variations depends on the level of the series [25]. A key feature of the Holt-Winters method is that it involves three smoothing parameters: α (level), β (trend), and γ (seasonality). These parameters are optimized to minimize the forecast error of the model [26].

Here, m denotes the length of the seasonal cycle, and S_{t-m} represents the seasonal component from the corresponding period in the previous cycle. Additionally, k refers to the integer part of $(h - 1)/m$, which ensures that the seasonal indices used in the forecasts correspond to those from the most recent seasonal cycle in the sample [27]. In Winters models, the seasonal components are normalized, meaning that the seasonal factors are adjusted so that their average equals 1. This implies that the sum of the seasonal components is equal to 1, ensuring that seasonal effects are appropriately scaled to the overall level of the series [28].

In summary, the four exponential smoothing methods differ mainly in the components they account for. The SES model considers only the level component and is suitable for series without trend or seasonality. The Holt method extends SES by including a trend component, allowing forecasts for data exhibiting a systematic increase or decrease. The Damped trend model further refines Holt's approach by introducing a damping parameter that gradually reduces the trend effect over time, providing more stable long-term forecasts. Finally, the Holt-Winters model incorporates both trend and seasonality, with additive and multiplicative formulations depending on whether seasonal fluctuations remain constant or vary in proportion to the series level.

Table 1. Holt-Winters equations (additive vs. multiplicative)

	Additive Model	Multiplicative Model
Forecast Equation	$\hat{y}_{t+h t} = \ell_t + hb_t + S_{t+h-m(k+1)}$	$\hat{y}_{t+h t} = (\ell_t + hb_t)S_{t+h-m(k+1)}$
Level Equation	$\ell_t = \alpha(y_t - S_{t-m}) + (1 - \alpha)(\ell_{t-1} + b_{t-1})$	$\ell_t = \alpha\left(\frac{y_t}{S_{t-m}}\right) + (1 - \alpha)(\ell_{t-1} + b_{t-1})$
Trend Equation	$b_t = \beta(\ell_t - \ell_{t-1}) + (1 - \beta)b_{t-1}$	$b_t = \beta(\ell_t - \ell_{t-1}) + (1 - \beta)b_{t-1}$
Seasonality Equation	$S_t = \gamma(y_t - \ell_{t-1} - b_{t-1}) + (1 - \gamma)S_{t-m}$	$S_t = \gamma\left(\frac{y_t}{\ell_{t-1} + b_{t-1}}\right) + (1 - \gamma)S_{t-m}$

Forecasting with Box-Jenkins Methods

Box-Jenkins models are considered a generalized form of exponential smoothing methods. While exponential smoothing focuses on modeling trend and seasonality components in the data, Box-Jenkins models aim to capture the autocorrelation structure within the series. These models are categorized as either seasonal or non-seasonal. Non-seasonal Box-Jenkins models are expressed as ARIMA(p,d,q), where p represents the order of the autoregressive (AR) component, d denotes the degree of differencing required to

achieve stationarity, and q indicates the order of the moving average (MA) component. Seasonal Box-Jenkins models are expressed as SARIMA(p,d,q)(P,D,Q)_m, where P , D , and Q represent the seasonal autoregressive, seasonal differencing, and seasonal moving average orders, respectively, and m denotes the seasonal period.

For Box-Jenkins models to be applicable, the time series must be stationary, meaning it should be free from trend and seasonal fluctuations. To achieve stationarity in the mean, differencing is applied, while variance stationarity is typically addressed using logarithmic or power transformations [9].

Non-seasonal Box-Jenkins models

Non-seasonal Box-Jenkins models include the autoregressive model (AR(p)), the moving average model (MA(q)), and their combination, ARMA(p,q). For non-seasonal series, if the series is not stationary (i.e., exhibits a trend), differencing is applied to achieve stationarity. In such cases, an integrated component I(d) is added to the model, resulting in the general form ARIMA(p,d,q).

p-th Order Autoregressive Model, AR(p)

Assuming the series is stationary, the general form of a p-th order autoregressive process (AR(p)) can be expressed as:

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \dots + \phi_p y_{t-p} + \varepsilon_t$$

where ε_t represents the white noise error term.

q-th Order Moving Average Model, MA(q)

The general form of a q-th order moving average process (MA(q)) is:

$$y_t = \varepsilon_t - \theta_1 \varepsilon_{t-1} - \theta_2 \varepsilon_{t-2} - \dots - \theta_q \varepsilon_{t-q}$$

ARMA(p,q) Model: Autoregressive (p) and Moving Average (q) Components

For a stationary series, the ARMA(p,q) model combines AR(p) and MA(q) components as:

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \dots + \phi_p y_{t-p} + \varepsilon_t - \theta_1 \varepsilon_{t-1} - \theta_2 \varepsilon_{t-2} - \dots - \theta_q \varepsilon_{t-q}$$

Here, p and q represent the orders of the AR and MA components, respectively, and d denotes the number of differences required for stationarity. To determine appropriate values for p and q , the autocorrelation function (ACF) and partial autocorrelation function (PACF) plots are widely used [6].

Seasonal Box-Jenkins

Seasonal variations can be effectively modeled using the SARIMA approach proposed within the Box-Jenkins framework. This model extends the standard ARIMA structure by incorporating seasonal components and is represented as SARIMA(p, d, q)(P, D, Q)_s. In this notation, p , d , and q denote the non-seasonal autoregressive (AR) order, the number of non-seasonal differences, and the non-seasonal moving average (MA) order, respectively; while P , D , and Q represent the seasonal AR, seasonal differencing, and seasonal MA orders, and s indicates the length of the seasonal period [29]. The general mathematical form of the SARIMA(p, d, q)(P, D, Q)_s model is as follows [30]:

$$(1 - \phi_1 B - \phi_2 B^2 - \dots - \phi_p B^p)(1 - \Phi_1 B^s - \Phi_2 B^{2s} - \dots - \Phi_P B^{sP})(1 - B)^d (1 - B^s)^D y_t = (1 - \theta_1 B - \theta_2 B^2 - \dots - \theta_q B^q)(1 - \Theta_1 B^s - \Theta_2 B^{2s} - \dots - \Theta_Q B^{sQ}) \varepsilon_t$$

In order to evaluate the forecasting performance of the models described above on the relevant dataset, several error metrics will be applied. The key performance indicators used for this purpose are presented below. Let y_t denote the actual observed value at time t , and $\hat{y}_t$ denote the forecasted value at time t .

Root Mean Square Error (RMSE):

RMSE is a frequently used statistic that calculates the average squared difference between the actual and anticipated values, squared as a root. Better model performance in terms of prediction accuracy is shown by lower RMSE, values:

$$RMSE = \sqrt{\frac{\sum_{t=1}^T (y_t - \hat{y}_t)^2}{T}}$$

Mean Absolute Error (MAE):

$$MAE = \frac{\sum_{t=1}^T |y_t - \hat{y}_t|}{T}$$

Mean Absolute Percentage Error (MAPE)

$$MAPE = \frac{100}{T} \sum_{t=1}^T \left| \frac{y_t - \hat{y}_t}{y_t} \right|$$

A low RMSE value indicates that the model is less sensitive to large deviations and demonstrates a higher overall forecasting performance. Similarly, a low MAE value suggests that the model's predictions are generally close to the actual values and reflect consistent predictive power. A low MAPE percentage shows that the predictions are relatively close to the actual values, thus indicating higher accuracy. When these three-error metrics are considered together, they provide a comprehensive assessment of the model's absolute and relative accuracy.

RESULTS AND DISCUSSION

In this study, air quality data for Çanakkale were obtained from continuous monitoring stations under the Marmara Clean Air Center Directorate. The study covers the period from January 2019 to June 2024. The raw data were collected on a daily basis and converted into monthly averages for analysis and modeling. The air pollutants considered in the study include PM₁₀, PM_{2.5}, SO₂, NO₂, and O₃. Time series plots and boxplots for these pollutants are presented in Figure 1.

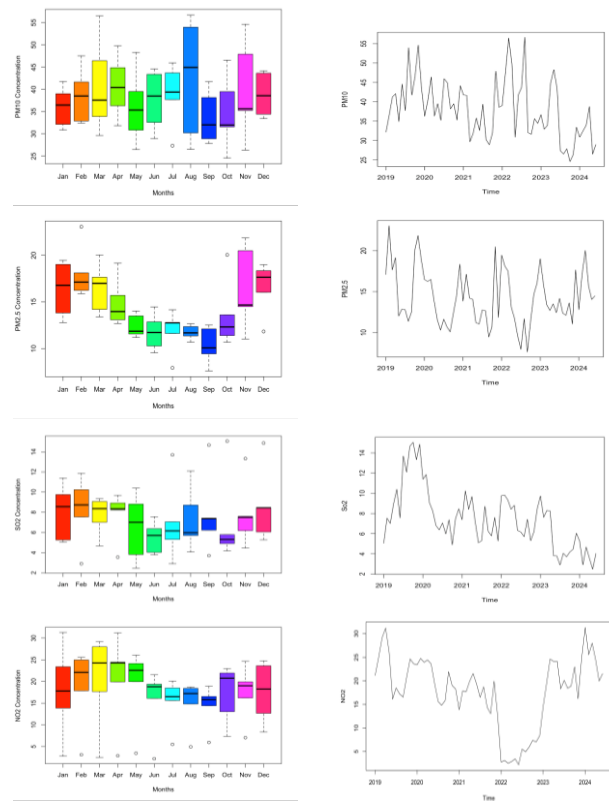


Figure 1. Monthly boxplots and time series plots of air pollutants observed in Çanakkale (2019–2024)

Among the air pollutants examined over the 2019–2024 period, the most evident trend structure was observed in the SO₂ series. SO₂ levels exhibited a steady decline over time, reaching minimum values particularly in 2023 and 2024, indicating a potential reduction in emissions from heating and industrial activities. On the other hand, the most prominent seasonal pattern was identified in the O₃ series. Ozone concentrations showed a regular increase during the summer months each year, reflecting the influence of photochemical reactions driven by sunlight (Figure 1).

Table 2 presents the basic descriptive statistics for the observed air pollutants in Çanakkale for the period 2019–2024. The results are compared with the threshold values specified in the Turkish Air Quality Assessment and Management Regulation. PM₁₀ concentrations ranged between 24.556 µg/m³ and 56.668 µg/m³, with an average value of 38.171 µg/m³. This value is very close to the annual limit (40 µg/m³) specified in the regulation, indicating that

the threshold was exceeded during certain periods. For PM_{2.5}, the mean concentration was 14.270 µg/m³, generally remaining below the limit value of 20 µg/m³. SO₂ concentrations ranged from 2.466 µg/m³ to 15.058 µg/m³, with an average of 7.476 µg/m³. These results demonstrate that SO₂ levels remained well below both annual and daily limit values. The average NO₂ concentration was 17.782 µg/m³, with a maximum of 31.309 µg/m³, which is below the annual limit of 42 µg/m³; however, periodic increases related to traffic and heating sources are noteworthy. For O₃, the average value was 51.259 µg/m³, while the maximum was recorded as 84.928 µg/m³, remaining below the 8-hour limit (120 µg/m³) defined by Turkish regulations. To identify the relationships among pollutants and uncover potential common source structures, factor analysis was applied. The analysis employed the principal components method, and factor loadings were made interpretable using varimax rotation.

Table 2. Descriptive statistics for observed air pollutants of Çanakkale city

Threshold Values of the Turkish Air Quality Directive		Min.	Max.	Mean	Std. Err. of Mean	95% Confidence Interval Mean	
						Lower	Upper
PM ₁₀	40 ^a µg/m. ³ - 50 ^b µg/m. ³	24.556	56.668	38.171	0.942	36.290	40.052
PM _{2.5}	20 ^d µg/m. ³	7.610	23.059	14.270	0.420	13.431	15.109
SO ₂	20 ^a µg/m. ³ - 125 ^b µg/m. ³	2.466	15.058	7.476	0.367	6.743	8.209
NO ₂	42 ^a µg/m. ³	2.196	31.309	17.782	0.919	15.947	19.618
O ₃	120 ^c µg/m. ³	20.034	84.928	51.259	2.061	47.143	55.374

a: annual, b: 24-hour, c: 8-hour, d: annual according to EU Directive

Prior to conducting the factor analysis presented in Table 3, the Kaiser-Meyer-Olkin (KMO) test and Bartlett's test of sphericity were applied to assess the suitability of the data for factor analysis. The KMO value was found to be 0.531, indicating a borderline adequacy for factor analysis. Bartlett's test of sphericity yielded $\chi^2 = 57.876$ with $p < 0.01$, confirming the presence of statistically significant correlations among variables and supporting the appropriateness of factor analysis. The factor analysis resulted in two components explaining approximately 64.3% of the total variance. The first component (Component 1) exhibited high loadings for PM₁₀ (0.85) and SO₂ (0.81), which can be interpreted as pollutants associated with residential heating and industrial activities. The second component (Component 2) had positive loadings for PM_{2.5} (0.81) and NO₂ (0.61), while O₃ showed a strong negative loading (-0.73). This pattern suggests that the second factor represents pollutants related to traffic emissions and photochemical processes. These findings indicate that air pollution in Çanakkale is influenced by multiple sources. The dominance of PM₁₀ and SO₂ during winter months supports the impact of heating and industrial emissions, whereas the prominence of O₃ in summer reflects the influence of photochemical reactions. The co-occurrence of NO₂ and PM_{2.5} within the same factor highlights the significance of traffic-related emissions. Overall, the factor analysis results reveal that PM₁₀ and SO₂ play the most prominent role in Çanakkale's air pollution. Their high loadings on the same factor emphasize shared source effects, particularly from heating and industrial activities. Therefore, the forecasting models in this study were developed based on PM₁₀ and SO₂.

Table 3. Factor analysis results

	Component 1	Component 2
PM ₁₀	0.85035	
SO ₂	0.81359	
PM _{2.5}		0.80945
NO ₂		0.60821
O ₃		-0.73299
% of Variance	%32.319	%31.996

A heatmap illustrating the number of days these two pollutants exceeded the limit values specified in the Turkish Air Quality Regulation is presented in Figure 2.

As shown in Figure 2, during the period 2019–2024, the total number of days in which PM₁₀ concentrations in Çanakkale exceeded the limit value specified in the Turkish Air Quality Regulation was 679, while for SO₂ it was 54. For PM₁₀, the highest exceedance occurred in 2019 (157 days) and 2020 (156 days). A notable decrease was observed in 2021 (93 days), followed by an increase in 2022 (133 days). In 2023 and 2024, a downward trend in PM₁₀ exceedance days became evident. Regarding SO₂, the highest exceedance was recorded in 2019 (35 days), while in subsequent years the frequency of exceedance declined significantly, with no exceedance observed in 2024.

In time series modeling, it is crucial to determine whether the series is stationary to ensure that the assumptions of the applied methods are satisfied. Stationarity is defined as the condition in which the mean and variance of the series remain constant over time, and the autocovariances depend only on

the lag length. When working with a non-stationary time series, regression analysis involving such series may lead to the problem of spurious or meaningless regression [31]. Although stationarity can be preliminarily assessed through graphical analysis, more advanced detection methods involve the use of autocorrelation functions (ACF), partial autocorrelation functions (PACF), and unit root tests (such as the Augmented Dickey-Fuller test). In this study, stationarity analysis was performed using the ADF test. The hypotheses for the ADF test were formulated as follows: H_0 : The series is non-stationary, H_1 : The series is stationary. Failure to reject the null hypothesis (when $p \geq 0.05$ at a Type I error level of 0.05) indicates that the series is non-stationary. The results of the ADF test are presented in Table 4.

According to the ADF test results, the PM_{10} series was found to be non-stationary ($p \geq 0.05$), whereas the SO_2 series was considered stationary at the 95% confidence level, as its p -

value was below 0.05 (Table 4). For the non-stationary series, first-order differencing was applied to achieve stationarity, and the transformed series were then incorporated into the modeling stage.

Following the stationarity analysis, time series forecasting was performed for PM_{10} and SO_2 , which play the most dominant role in air pollution in Çanakkale. For this purpose, two classical time series approaches were employed in a comparative framework: Exponential Smoothing (SES, Holt, Damped, Holt-Winters additive and multiplicative) and the Box-Jenkins methodology (non-seasonal ARIMA and seasonal SARIMA). Forecasts were generated for both in-sample (January 2019–June 2024) and out-of-sample (July–December 2024) periods, and model performance was evaluated using RMSE, MAE, and MAPE. The resulting best-performing models and their parameter specifications for PM_{10} and SO_2 are summarized in Tables 5-8.

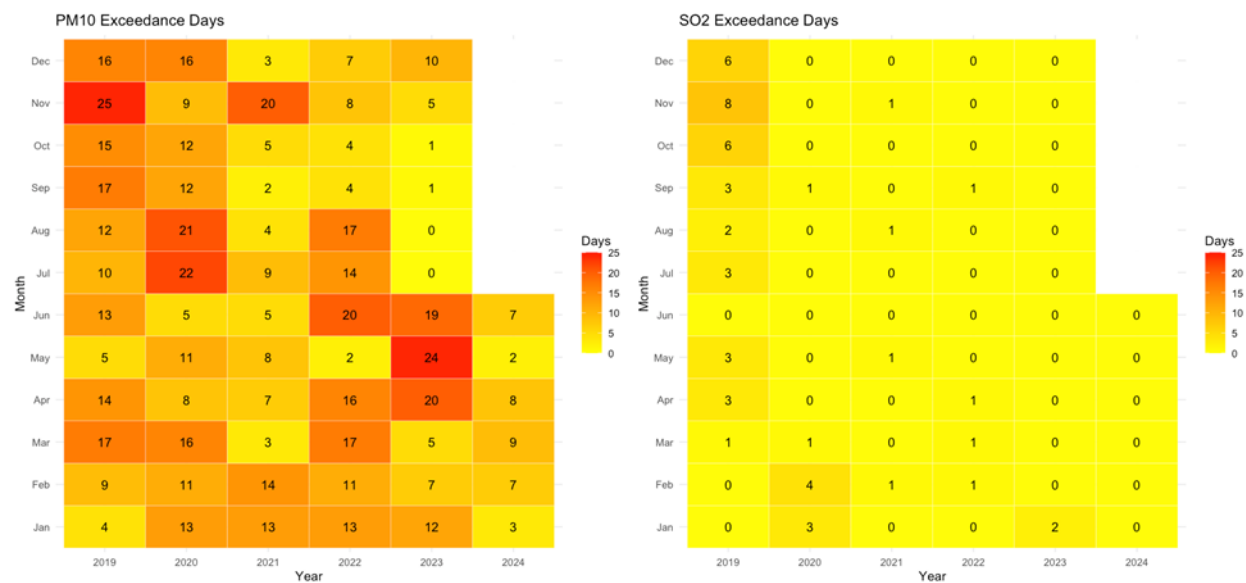


Figure 2. Heatmap of PM_{10} and SO_2 daily limit exceedances

Table 4. ADF unit root test results

Variable	Test Statistic	p-value	Decision	Differencing Order
PM_{10}	-2.992	0.173	Non-stationary	1
SO_2	-3.779	0.025	Stationary	0

Table 5. Comparison of forecasting performance of ETS and Box-Jenkins models for PM_{10}

	PM_{10}	RMSE	MAE	MAPE	Optimal Parameter
ETS	SES	7.105	5.646	15.295	$\alpha = 0.287$
	Holt	7.095	5.633	15.126	$\alpha = 0.271, \beta = 0.0004$
	Damped	7.048	5.670	15.472	$\alpha = 0.264, \beta = 0.0002, \theta = 0.8$
	Holt-Winter (Add.)	6.608	5.384	14.613	$\alpha = 0.297, \beta = 0.0001, \gamma = 0.0001$
	Holt-Winter (Mult.)	6.525	5.207	14.073	$\alpha = 0.443, \beta = 0.0007, \gamma = 0.0001$
Box-Jenkins	ARIMA(1,1,1)	6.847	5.413	14.658	
	SARIMA	6.847	5.413	14.658	

According to the error metrics presented in Table 5, the Holt-Winters methods, particularly the model incorporating a multiplicative seasonal component, demonstrated the best performance in forecasting PM_{10} concentrations. This model achieved lower error rates by effectively capturing both trend and seasonality. The additive Holt-Winters model ranked as the second-best performing method. In contrast, the SES model, which accounts only for the level component, produced higher error values compared to the other approaches. These findings highlight the significant role of the seasonal component in forecasting the PM_{10} series. However, when evaluating the Box-Jenkins methodology, no performance difference was observed between the ARIMA(1,1,1) and SARIMA models. This indicates that the seasonal fluctuations in the PM_{10} series are not statistically strong or consistent enough to improve forecasting accuracy through the addition of seasonal terms. ETS methods can flexibly incorporate trend and seasonality as separate components, which allows them to capture even weak seasonal effects, leading to better predictive performance. In contrast, within the Box-Jenkins framework, seasonality contributes to the model only if it is pronounced and regular; therefore, the SARIMA model did not provide additional benefits over the non-seasonal ARIMA model.

After examining the results obtained for the PM_{10} series in the in-sample period (January 2019–June 2024) presented in

Table 5, the analyses conducted to evaluate the performance of the models on the end-of-period data (July 2024–December 2024) are presented in Table 6.

According to the out-of-sample forecasting results presented in Table 6, the lowest error values for the PM_{10} series were obtained from the Holt-Winters (multiplicative) model. This finding indicates that the model maintained its superior performance in both the in-sample and out-of-sample periods by effectively capturing trend and seasonality components. The Holt-Winters (additive) model achieved the second-best performance, further confirming the substantial contribution of modeling seasonality to forecasting accuracy. In contrast, the seasonal fluctuations in the PM_{10} series are not sufficiently strong or consistent within the Box-Jenkins framework to meaningfully enhance forecasting accuracy.

Following the out-of-sample forecasting results presented in Table 6, Figure 3 illustrates the forecasts obtained from the Holt-Winters (multiplicative) model for the PM_{10} series, along with the 95% confidence intervals and the actual values for the corresponding period.

Figure 3 presents the PM_{10} forecasts obtained using the Holt-Winters (multiplicative) model for the period July–December 2024, along with the 80% (dark blue) and 95% (light blue) confidence intervals. These intervals reflect the uncertainty range of the model's predictions, with the actual observed values largely falling within these bounds.

Table 6. Out-of-sample forecasting performance of ETS and Box-Jenkins models for PM_{10}

	PM_{10}	RMSE	MAE	MAPE
ETS	SES	4.789	3.999	5.963
	Holt	4.275	3.289	13.337
	Damped	4.869	4.094	16.320
	Hold-Winter (Add.)	4.245	3.822	14.873
	Holt-Winter (Mult.)	3.275	2.823	11.040
Box-Jenkins	ARIMA(1,1,1)	6.352	5.940	23.071
	SARIMA	6.352	5.940	23.071

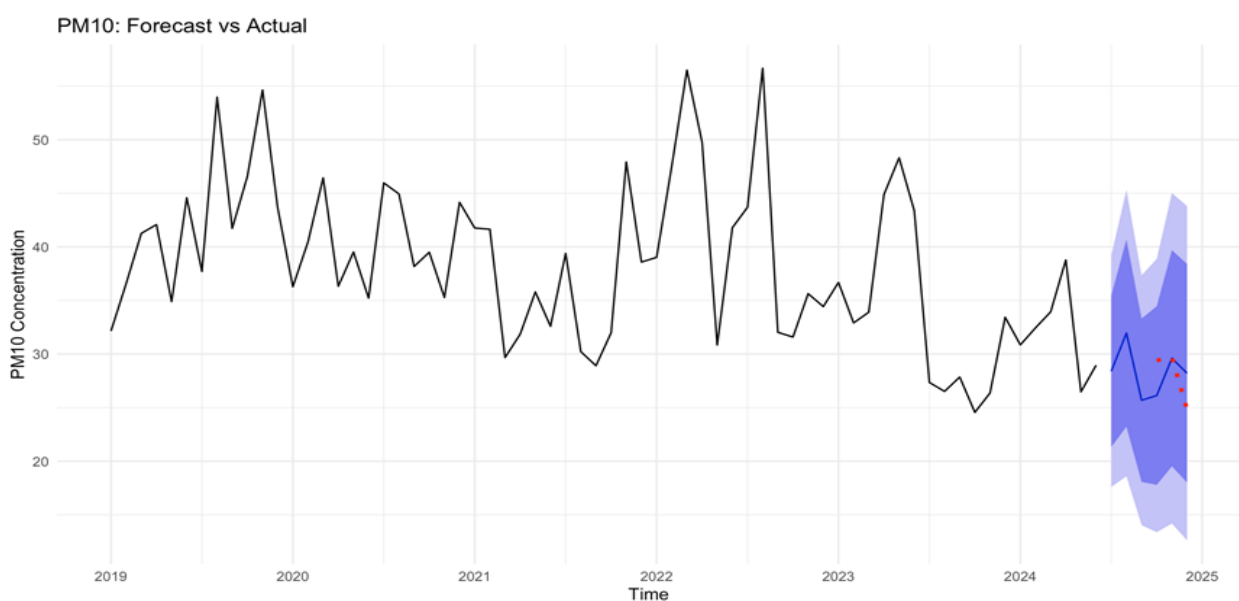


Figure 3. PM_{10} forecast vs actual values using Holt-Winters (Multiplicative) model

According to the error metrics presented in Table 7, among the ETS methods, the Holt-Winters (additive) model demonstrated the best performance for the SO₂ series. This model achieved the lowest RMSE (1.731) and MAPE (20.15%) values by accounting for both trend and weak seasonality effects. Although the Holt-Winters (multiplicative) model exhibited similar performance, its error rates were higher compared to the additive model. In contrast, the SES model, which only incorporates the level component, produced the highest error values among the ETS methods. For the Box-Jenkins methodology, the ARIMA(1,1,0) and SARIMA models yielded identical performance metrics. This indicates that the SO₂ series does not exhibit a strong seasonal pattern and adding seasonal components to the model does not improve forecasting accuracy. Furthermore, although the ADF test

suggested that the SO₂ series was stationary (Table 4), the ARIMA(1,1,0) model, selected based on information criteria, showed that applying first-order differencing slightly improved model fit. Overall, ETS methods outperformed the Box-Jenkins approach, and the Holt-Winters method, which accounts for trend and weak seasonality components, was identified as the most appropriate model for forecasting the SO₂ series.

Following the in-sample (January 2019–June 2024) forecasting results for the SO₂ series presented in Table 7, the analyses evaluating the models' out-of-sample (July 2024–December 2024) performance are provided in Table 8.

Table 7. In-sample forecasting performance of ETS and Box-Jenkins models for SO₂

	SO ₂	RMSE	MAE	MAPE	Optimal Parameter
ETS	SES	1.832	1.489	22.596	$\alpha = 0.688$
	Holt	1.834	1.485	22.365	$\alpha = 0.671, \beta = 0.0001$
	Damped	1.780	1.445	22.277	$\alpha = 0.648, \beta = 0.0001, \theta = 0.8$
	Holt-Winter (Add.)	1.731	1.339	20.154	$\alpha = 0.651, \beta = 0.0001, \gamma = 0.0001$
	Holt-Winter (Mult.)	1.838	1.436	22.105	$\alpha = 0.581, \beta = 0.0017, \gamma = 0.0762$
Box-Jenkins	ARIMA(1,1,0)	1.799	1.469	22.184	
	SARIMA	1.799	1.469	22.184	

Table 8. Out-of-sample forecasting performance of ETS and Box-Jenkins models for SO₂

	SO ₂	RMSE	MAE	MAPE
ETS	SES	1.535	1.250	23.107
	Holt	1.703	1.361	24.844
	Damped	1.552	1.258	23.174
	Hold-Winter (Add.)	1.768	1.412	26.447
	Holt-Winter (Mult.)	1.241	1.048	20.253
Box-Jenkins	ARIMA(1,1,0)	1.593	1.263	22.875
	SARIMA	1.593	1.263	22.875

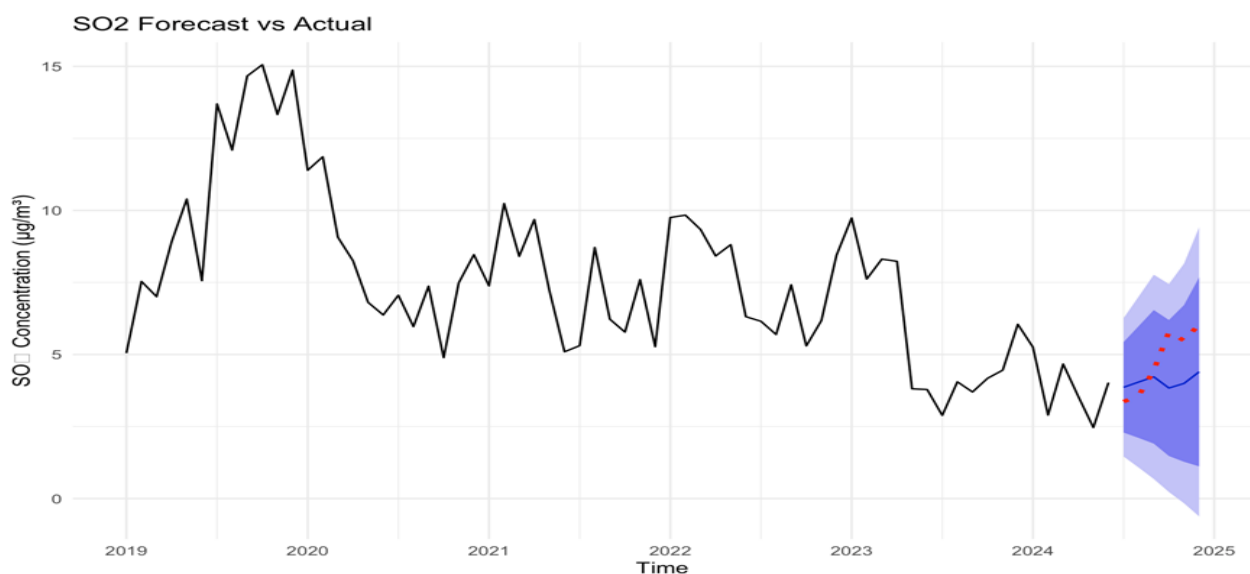


Figure 4. SO₂ forecast vs actual values using Holt-Winters (Multiplicative) model

An examination of the out-of-sample results presented in Table 8 reveals that the lowest error values for the SO₂ series were obtained from the Holt-Winters (multiplicative) model. While the Holt-Winters (additive) model produced better results during the in-sample period, the multiplicative model outperformed it in the out-of-sample period primarily due to changes in the amplitude of the seasonal fluctuations. The additive model assumes that seasonal effects remain constant in magnitude, whereas the multiplicative model allows these effects to vary in proportion to the series level. During the test period, the increase in SO₂ levels was accompanied by a proportional increase in the magnitude of seasonal fluctuations. Consequently, the multiplicative model was better able to capture this changing amplitude pattern, resulting in lower error values. Therefore, the Holt-Winters (multiplicative) model, which effectively accounts for both trend and proportional seasonality components, was ultimately identified as the most suitable model for forecasting the SO₂ series.

Following the out-of-sample forecasting performance results presented in Table 8, Figure 4 illustrates the forecasts obtained from the Holt-Winters (multiplicative) model for the SO₂ series, along with the 80% and 95% confidence intervals, in comparison with the actual values for the corresponding period.

An examination of Figure 4 reveals that the values forecasted by the Holt-Winters (multiplicative) model align closely with the actual SO₂ observations for the July–December 2024 period. In the graph, the light blue area represents the 95% confidence interval, while the dark blue area corresponds to the 85% confidence interval. The forecasts generally fall within these bounds, indicating that the model's predictions are statistically reliable.

Overall, while direct numerical comparison with previous studies such as Temiz and Turgut [11] is not feasible due to differences in data scope and modeling frameworks, the present findings conceptually complement earlier Box-Jenkins-based research by expanding the analysis to multiple pollutants and confirming the consistent superiority of the Holt-Winters multiplicative model in capturing both trend and proportional seasonality. Moreover, the results are in line with those of Cekim [12], who compared ETS and ARIMA models for PM₁₀ forecasting across Turkey's most polluted cities and found that ETS models could outperform ARIMA under certain urban conditions. In this sense, the present study contributes uniquely to the Turkish air quality literature by providing a detailed ETS–Box-Jenkins comparison for both PM₁₀ and SO₂ specifically in the Çanakkale region, thereby offering a region-focused statistical framework that can serve as a reference for future local air pollution forecasting studies.

CONCLUSIONS

This study conducted a comprehensive comparison of Exponential Smoothing and Box-Jenkins models for forecasting air pollution in Çanakkale, focusing on PM₁₀ and SO₂, which were identified as the most influential pollutants through factor analysis. Stationarity tests revealed that PM₁₀ required first-order differencing, while SO₂ was stationary at level. The forecasting results demonstrated that ETS methods generally outperformed Box-Jenkins models in terms of predictive accuracy. For PM₁₀, the Holt-Winters multiplicative model achieved the lowest RMSE and MAPE values, effectively capturing both trend and seasonal variations. Similarly, for SO₂, the Holt-Winters multiplicative

model provided the best performance, indicating that even weak seasonal components, when modeled appropriately, enhance forecast accuracy. Conversely, the Box-Jenkins methodology showed no difference between ARIMA and SARIMA models, highlighting the limited role of seasonality in these series. Overall, the findings emphasize that ETS methods offer greater flexibility in incorporating trend and seasonal patterns, making them superior for forecasting air pollutant concentrations in regions with complex temporal dynamics.

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DATA AVAILABILITY STATEMENT

The authors confirm that the data that supports the findings of this study are available within the article. Raw data that support the finding of this study are available from the corresponding author, upon reasonable request.

CONFLICT OF INTEREST

The authors declare that there is no conflict of interest with any individual, institution, or organization in the preparation, evaluation, or publication of this study.

USE OF AI FOR WRITING ASSISTANCE

Not declared.

ETHICS

There are no ethical issues with the publication of this manuscript.

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